Yalda Foroutan

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Education

Ph.D. in Computing Science *(Jan 2024 - Current)*

Simon Fraser University (SFU)

Supervisior: Andrea Tagliasacchi
Courses: 3D Vision, Computer Vision

Ph.D. in Engineering Science *(Sep 2021 - Apr 2023)*

Simon Fraser University (SFU)

Supervisior: Ivan V. Bajić
Courses: Information Theory, Deep Learning Systems, Stochastic Systems, Linear System Theory, Multimedia Communications, Digital Image Processing

M.Sc. in Electrical Engineering *(Sep 2017 - Sep 2020)*

University of Tehran

Thesis: Control of Computer's Mouse Using Hand Gesture Recognition
Supervisiors: Ahmad Kalhor, Samad Sheikhaei

B.Sc. in Electrical Engineering (Control) *(Sep 2012 - Sep 2017)*

Amirkabir University of Technology (Tehran Polytechnic)

Thesis: Driver's Consciousness Level Analysis Using EEG Signals
Supervisior: Mohammad A. Khosravi

Online Courses

Coursera

Deep Neural Networks with PyTorch, Neural Networks and Deep Learning, Improving Deep Neural Networks
Structuring Machine Learning Projects, Convolutional Neural Networks, Introduction to Calculus
Convolutional Neural Networks in TensorFlow, Introduction to TensorFlow, Sequence Models, Machine Learning

Research Experience

Research Assistant at GrUVi Lab *(Jan 2024 - Current)*

Simon Fraser University

Developed a NeRF-guided training procedure for Gaussian Splatting, eliminating the need for SfM initialization and enhancing scene reconstruction quality for complex and dynamic environments
Under Supervision of Andrea Tagliasacchi
Focused Expertise: 3D Vision, Novel View Synthesis

Research Assistant at Multimedia Lab *(Feb 2022 - Apr 2023)*

Simon Fraser University

Developed scalable image codecs for human and machines, primarily optimizing the efficiency of base and enhancement layers
Contributed significantly to the development of four research papers
Under Supervision of: Ivan V. Bajić
Focused Expertise: Learnable Image Compression, Compressed Vision, Image and Video Coding

Research Assistant at Advanced Circuits for Data Communication Lab *(Feb 2018 - Sep 2020)*

University of Tehran

Developed hand gesture recognition application and collected a hand dataset containing four gestures
Under Supervision of: Ahmad Kalhor & Samad Sheikhaei
Focused Expertise: Hand Gesture Recognition, Object Detection, Image Classification

Publications

Evaluating Alternatives to SfM Point Cloud Initialization for Gaussian Splatting
Yalda Foroutan, Daniel Rebain, Kwang Moo Yi, Andrea Tagliasacchi
Under Submission

Base Layer Efficiency in Scalable Human-Machine Coding
Yalda Foroutan, Alon Harell, Anderson Andrade, Ivan V. Bajic
IEEE International Conference on Image Processing, ICIP 2023, Kuala Lumpur, Malaysia.

Rate-Distortion Theory in Coding for Machines and its Application
Alon Harell, Yalda Foroutan, Nilesh A. Ahuja, Parual Datta, Bhavya Kanzariya, V. Srinivasa Somayazulu, Omesh Tickoo, Anderson Andrade, Ivan V. Bajic

Submitted in IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI).

VVC+M: Plug and Play Scalable Image Coding for Humans and Machines
Alon Harell, Yalda Foroutan, Ivan V. Bajic
IEEE International Conference on Multimedia and Expo Workshops, ICMEW Workshops 2023, Brisbane, Australia.

Conditional and Residual Methods in Scalable Coding for Humans and Machines
Anderson Andrade, Alon Harell, Yalda Foroutan, Ivan V. Bajic
IEEE International Conference on Multimedia and Expo Workshops, ICMEW Workshops 2023, Brisbane, Australia.

Control of Computer Pointer using Hand Gesture Recognition in Motion Pictures
Yalda Foroutan, Ahmad Kalhor, Saeid Mohammadi Nejati, Samad Sheikhaei
arXiv 2020.

Academic Projects

3D Vision (CMPT895) *SFU*

Course Mini-projects

Moving Least Squares (MLS)
3D Morphable Face Model; Rigid Shape Matching, Rigid Iterative Closest Point (ICP), and Non-Rigid Fitting
Neural Field Reconstruction

Computer Vision (CMPT762) *SFU*

Course Mini-projects

Digit Recognition with Convolutional Neural Networks
MNIST Digit Recognition with Convolutional Neural Networks
Optimizing CIFAR-100 Image Classification
Detection, Semantic Segmentation, Instance Segmentation
3D Reconstruction

Deep Learning (ENSC813) *SFU*

Course Final Project

Finding the Minimum Bitrates for a Computer Vision Task

Multimedia Communication (ENSC424) *SFU*

Course Final Project

Evaluating the Efficiency and Limitations of CANF-VC: Enhancing Performance for Shot Transitions, Motion Information, and Reference Frames

Information Theory (ENSC808) *SFU*

Seminar

Presentation and Proving Equations of Information Bottleneck Paper

Digital Image Processing (ENSC895) *SFU*

Course Final Project

Segmentation of Breast Cancer Imaging

Neural Networks and Deep Learning *University of Tehran*

Course Mini-projects

Design of a CNN Classifier for Fashion MNIST Dataset
Implementation of RNN for Stock Market Prediction
Text Generation Based on Shakespeare's Book with RNN Networks
DCGAN, WGAN, and ACGAN Implementations on Fashion MNIST and CIFAR-10 Datasets

Other Python Projects

Independent Projects

Face Filters: Changing Eye Color, Eye Size, and Putting 3D Sunglasses
Enhancing Low-Light Images and Increasing Image Resolution using GANs
Low light Image Enhancement using GANs
Increasing Image Resolution

Skills

Programming Languages Python, MATLAB, C/C++, Julia
Frameworks and Libraries PyTorch, TensorFlow, OpenCV
Other Tools Bash, HTML/CSS, R

Teaching Experience

2025	**Computer Vision (CMPT762)**	*SFU*
2024	**Introduction to Engineering Analysis (ENSC180)**	*SFU*
2023	**Capstone B (ENSC440)**	*SFU*
2023	**Graphical Communication (ENSC204)**	*SFU*
2023	**Capstone A (ENSC405)**	*SFU*
2023	**Computational Neuroscience** (Certificate)	*Neuromatch Academy*
2023	**Capstone A (ENSC405)**	*SFU*
2022	**Graphical Communication (ENSC204)**	*SFU*
2021	**Computational Neuroscience** (Certificate)	*Neuromatch Academy*
2021	**Deep Learning** (Certificate)	*Neuromatch Academy*
2020	**Neural Networks and Deep Learning**	*University of Tehran*
2020	**Electronic 1**	*University of Tehran*
2019	**Electronic 1**	*University of Tehran*
2019	**Electronic 3**	*University of Tehran*

Awards and Honors

2024	**CMPT Graduate Fellowships** from Computing Science department (CMPT) at SFU	*Canada*
2024	**Graduate Fellowship** from SFU	*Canada*
2024	**PhD Research Scholarship** from SFU	*Canada*
2022	**Talent Bursary** from Alberta Machine Intelligence Institute (amii)	*Canada*
2020	**Ph.D. Admission** from University of Tehran	*Iran*
2008	**2nd Place** in Iran Mathematics Olympiad PAYA	*Iran*
2006	**1st Place** in Abadan Mathematics Olympiad	*Iran*
2005	**Exceptional Talent Recognition** by National Organization for Development of Exceptional Talents (NODET)	*Iran*

The electronic version contains hyperlinks.
For the extended version and more information, feel free to visit https://youlenda.github.io